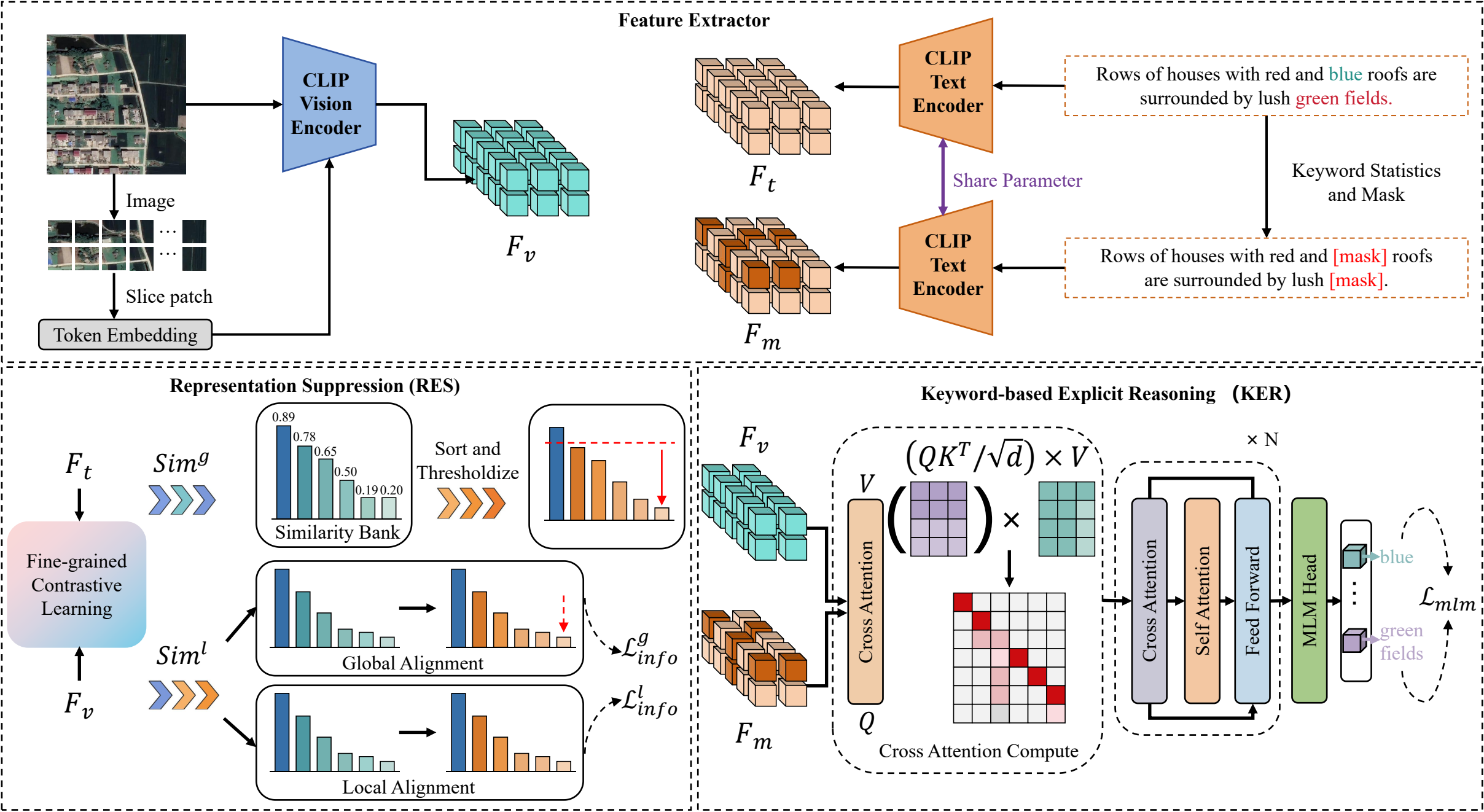

Feature Extractor
CLIP Vision Encoder
Image
Slice patch
Token Embedding
F_v
CLIP Text Encoder
Rows of houses with red and blue roofs are surrounded by lush green fields.
F_t
Share Parameter
Keyword Statistics and Mask
CLIP Text Encoder
Rows of houses with red and [mask] roofs are surrounded by lush [mask].
F_m
Representation Suppression (RES)
F_t
Sim^g
0.89
0.78
0.65
0.50
0.19
0.20
Similarity Bank
Sort and Thresholdize
Fine-grained Contrastive Learning
Sim^l
F_v
Global Alignment
$\mathcal{L}^g_{info}$
Local Alignment
$\mathcal{L}^l_{info}$
Keyword-based Explicit Reasoning (KER)
F_v
F_m
V
Q
Cross Attention
$(QK^T/\sqrt{d}) \times V$
Cross Attention Compute
× N
Cross Attention
Self Attention
Feed Forward
MLM Head
blue
green fields
$\mathcal{L}_{mlm}$